A Course End Project

Smart Home Security System Using PIR Sensor

IOT LABORATORY (A8603)

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

SUBMITTED BY

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This is to certify that the Course End Project titled **"Smart Home Security System Using PIR Sensor"** is carried out by **A. Sahasra Reddy (24881A0566), A. LaxmiSri (24881A0569), and B. Shruthi (24881A0571)**, towards **A8603 – IOT Laboratory** course in partial fulfilment of the requirements for the award of degree of **Bachelor of Technology in Computer Science and Engineering** during the Academic Year 2025–26.

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Check List

S. No.	Content	Status (✓ / ✗)
1	Abstract	✓
2	Objectives	✓
3	Problem Statement	✓
4	Algorithm	✓
5	Flow Chart	✓
6	Source Code	✓
7	Input	✓
8	Output	✓
9	Conclusion	✓
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Table of Content

1 Abstract

This project titled **Smart Home Security System using PIR Sensor** focuses on enhancing home security through the application of sensor-based technology and automation. With the increasing need for safety in residential environments, the system aims to detect unauthorized access and provide immediate alerts, ensuring better protection and monitoring.

The proposed system utilizes a Passive Infrared (PIR) sensor to detect human motion by sensing changes in infrared radiation emitted by living beings. When motion is detected within the sensor's range, the system processes the signal and activates an alert mechanism such as a buzzer or LED indicator. This enables instant identification of intrusions and improves response time in critical situations.

The system is designed to operate automatically without requiring continuous human supervision. It can be installed at entry points such as doors, windows, or restricted areas to monitor movement effectively. The simplicity of the design ensures ease of installation and maintenance, making it suitable for both indoor and small-scale security applications.

From a computer science perspective, the project involves basic signal processing, sensor interfacing, and control mechanisms. It demonstrates the practical implementation of embedded system concepts and automation techniques in real-world applications. The system emphasizes reliability, low power consumption, and cost-effectiveness.

The Smart Home Security System is a simple yet efficient solution that enhances safety, reduces manual monitoring, and provides a foundation for further advancements such as integration with alarms, cameras, or wireless communication systems. It highlights the importance of automation and sensor-based technologies in developing secure and intelligent environments.

Keywords: PIR Sensor, Smart Home Security, Motion Detection, Intruder Detection, Alert System, Infrared Sensor, Automation, Embedded Systems, Home Safety, Security System.

2 Problem Statement

Ensuring safety and security in residential environments has become a major concern in recent times due to increasing cases of unauthorized access and theft. Traditional security methods, such as manual surveillance or basic alarm systems, are often inefficient, as they require constant human monitoring and may fail to provide immediate alerts during critical situations. These systems lack automation and real-time responsiveness, making them less reliable for modern security needs.

The problem becomes more significant in situations where homeowners are not physically present, as there is no effective mechanism to detect and alert intrusions instantly. Additionally, many conventional systems are expensive and complex, making them inaccessible for small-scale or low-cost applications. There is a need for a simple, cost-effective, and automated solution that can detect human presence and provide immediate alerts without requiring continuous supervision.

Therefore, there is a need for a smart home security system that can:

- Detect human motion accurately using a PIR sensor,
- Automatically trigger alert mechanisms such as buzzer or LED,
- Provide real-time indication of unauthorized entry,
- Operate with low power consumption and minimal cost, and
- Be easy to install and maintain in residential environments.

By implementing this system, homeowners can improve safety, reduce dependency on manual monitoring, and respond quickly to potential threats. The Smart Home Security System using a PIR sensor demonstrates how embedded systems and automation techniques can be applied to develop an efficient and reliable security solution for real-world applications.

3 Objectives

1. Intrusion Detection using PIR Sensor: To detect human motion accurately using a Passive Infrared (PIR) sensor for identifying unauthorized access in residential areas.
2. Automatic Alert Mechanism: To automatically activate alert systems such as a buzzer or LED when motion is detected, ensuring immediate notification of intrusions.
3. Real-Time Monitoring: To continuously monitor the surroundings and provide instant response without requiring human intervention.
4. Low Cost and Energy Efficiency: To design a cost-effective system that consumes minimal power and can operate efficiently for long durations.
5. Ease of Installation and Usage: To develop a simple system that can be easily installed in homes without requiring complex setup or technical expertise.
6. Practical Application of Embedded Systems: To demonstrate the implementation of sensor-based technology and automation techniques in building a real-world home security system.

4 Algorithm

1. **System Initialization:**
   (a) Power ON the system.
   (b) Initialize microcontroller (NodeMCU).
   (c) Initialize PIR sensor and output devices (buzzer/LED).
2. **Read Sensor Data:**
   (a) Continuously monitor the PIR sensor.
   (b) Detect changes in infrared radiation.
3. **Motion Detection:**
   (a) If motion is detected by the PIR sensor $\rightarrow$ intrusion detected.
   (b) If no motion $\rightarrow$ system remains in monitoring state.
4. **Alert Activation:**
   (a) Turn ON buzzer or LED when motion is detected.
   (b) Maintain alert for a specific duration.
5. **Reset System:**
   (a) Turn OFF buzzer/LED after alert duration.
   (b) Reset system to monitoring mode.
6. **Continuous Monitoring:**
   (a) Repeat sensor reading and detection process continuously.
   (b) Ensure real-time response to any movement.
7. **End:**
   (a) System runs continuously until manually powered OFF.

5 Proposed System

The proposed Smart Home Security System is designed to provide an efficient and automated solution for detecting unauthorized access in residential environments. The system continuously monitors the surroundings using a Passive Infrared (PIR) sensor, which detects human motion by sensing changes in infrared radiation emitted by living beings.

When motion is detected, the sensor sends a signal to the processing unit (microcontroller), which immediately activates an alert mechanism such as a buzzer or LED indicator. This ensures real-time notification of intrusions and allows users to take immediate action. The system operates automatically without the need for continuous human monitoring, making it reliable and efficient.

The proposed system is simple in design, cost-effective, and consumes minimal power, making it suitable for continuous operation in homes. It can be installed at entry points such as doors, windows, and restricted areas to enhance security coverage. The system is also easy to install and maintain, making it accessible for a wide range of users.

The design is flexible and can be extended in the future by integrating additional features such as alarms, cameras, or wireless communication for remote monitoring. By combining sensor-based detection and automation, the Smart Home Security System provides a practical and efficient solution for improving safety and security in modern homes.

5.1 Flow Chart

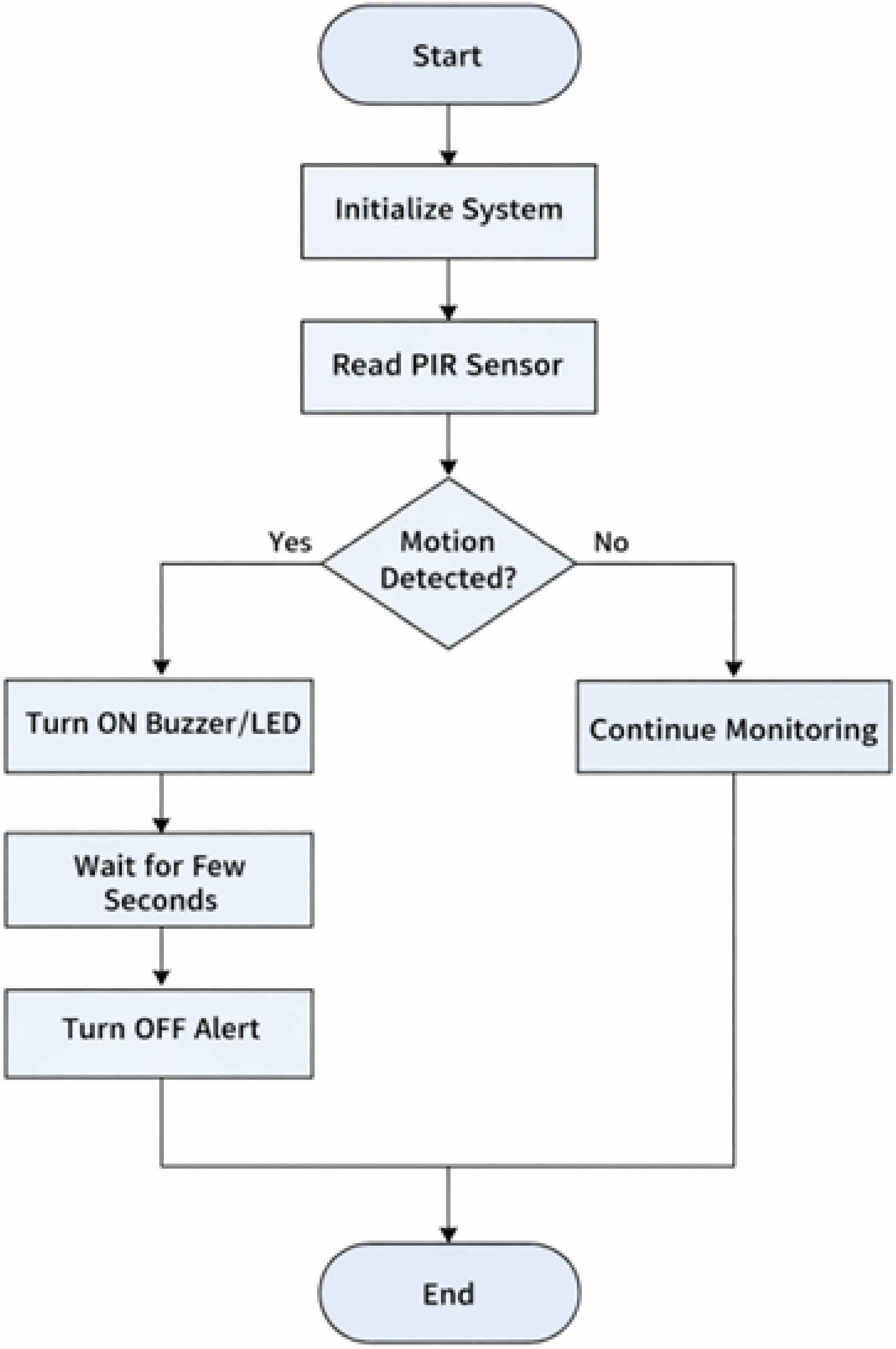


Figure 1: Flow Chart

5.2 Working Model

- **Objective:** To detect human motion using a PIR sensor and provide immediate alerts through a buzzer or LED, enhancing home security and reducing the need for continuous manual monitoring.
- **Components Required:**
  - Microcontroller: Arduino UNO or NodeMCU
  - Sensor:
    - PIR Sensor (detects human motion)
  - Output Devices:
    - Buzzer (for sound alert)
    - LED (for visual indication)
  - Power Supply (5V)
  - Connecting wires and breadboard
- **System Workflow:**
  - Initialization: Microcontroller powers ON and initializes PIR sensor and output devices.
  - Data Monitoring: PIR sensor continuously monitors infrared radiation changes.
  - Motion Detection:
    - If motion detected → Signal sent to microcontroller.
    - Else → System continues monitoring.
  - Alert Activation:
    - Buzzer and/or LED turns ON when motion is detected.
    - Alert continues for a few seconds.
  - Reset:
    - Buzzer/LED turns OFF after delay.
    - System returns to monitoring state.
  - Continuous Operation: Process repeats continuously for real-time security.
- **Implementation Steps:**
  - Connect PIR sensor and buzzer/LED to the microcontroller.
  - Write and upload program to read sensor input and control outputs.
  - Test the system by simulating human motion.

    - Adjust sensitivity and delay of PIR sensor if required.

- **Advantages of This Working Model:**
    - Automatic intrusion detection
    - Low cost and energy efficient
    - Easy to install and maintain
    - Real-time alert system
    - Scalable for future enhancements (IoT, cameras, alarms)

6 Source Code

```

// PERFECT WIRING - Final Production Code
#define PIR_PIN 4    // D2
#define BUZZER_PIN 5 // D1

bool alarmActive = false;
unsigned long alarmStart = 0;
const unsigned long ALARM_TIME = 10000; // 10 seconds

void setup() {
  Serial.begin(115200);
  pinMode(PIR_PIN, INPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW);

  Serial.println("Security System - Wiring PERFECT!");
  Serial.println("Calibrating PIR (30s)... STAY STILL!");

  // Startup beep
  for(int i = 0; i < 3; i++) {
    digitalWrite(BUZZER_PIN, HIGH);
    delay(100);
    digitalWrite(BUZZER_PIN, LOW);
    delay(200);
  }

  delay(30000); // PIR calibration
  Serial.println("SYSTEM ARMED - Ready for intruders!");
}

void loop() {
  bool pirState = digitalRead(PIR_PIN);

  if (pirState == HIGH && !alarmActive) {
    // MOTION DETECTED!
    alarmActive = true;
    alarmStart = millis();
    Serial.println(" INTRUDER DETECTED!");
    Serial.println("BUZZER ACTIVATED ON D1!");
  }
```

```

  // ALARM CONTROL
  if (alarmActive) {
    digitalWrite(BUZZER_PIN, HIGH);  // D1 HIGH = Buzzer ON

    if (millis() - alarmStart > ALARM_TIME) {
      digitalWrite(BUZZER_PIN, LOW);   // D1 LOW = Buzzer OFF
      alarmActive = false;
      Serial.println("System reset - ARMED");
    }
  }

  delay(50);
}
```

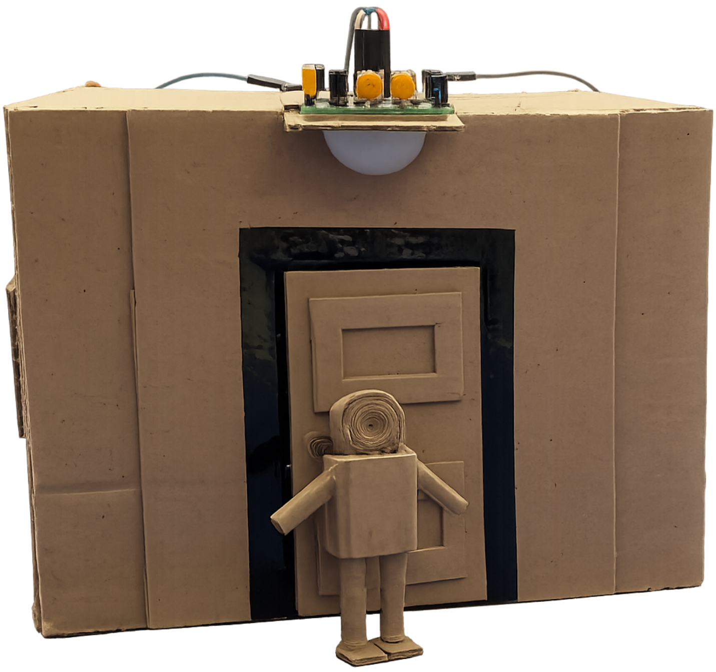
7 Input

The Smart Home Security System takes the following inputs to operate efficiently:

- **PIR Sensor Reading:** Detects motion by sensing changes in infrared radiation emitted by human bodies. This input determines whether an intruder is present or not.
- **System Status (Armed/Disarmed):** Controlled through the Blynk mobile application. When the system is armed, motion detection is active; when disarmed, alerts are disabled.
- **WiFi Connectivity:** Required for IoT functionality, enabling communication between the NodeMCU and the Blynk cloud for real-time monitoring and notifications.
- **User Commands (Optional):** Allows the user to control the system remotely (e.g., turning the system ON/OFF) through the mobile application.
- **Timing Control (Optional):** Defines how frequently the system checks sensor input and triggers alerts, ensuring stable and accurate motion detection.

These inputs enable the system to detect motion accurately, provide real-time alerts, and ensure efficient and automated home security monitoring.

8 Output



```
🔒 Security System - Wiring PERFECT!
Calibrating PIR (30s)... STAY STILL!
[30 seconds silence - startup beeps heard]
✅ SYSTEM ARMED - Ready for intruders!
[Silence continues...]
```

9 Conclusion

The Smart Home Security System using a PIR sensor effectively demonstrates the application of automation and embedded systems in enhancing residential safety. By continuously monitoring motion through infrared sensing, the system is capable of detecting unauthorized access in real-time and immediately triggering an alert mechanism such as a buzzer or LED. This ensures quick response to potential threats and improves overall security.

One of the key advantages of this system is its simplicity and reliability. The PIR sensor accurately detects human movement, and the microcontroller processes the input efficiently to activate alerts without delay. The system operates automatically without requiring constant human supervision, making it a cost-effective and user-friendly solution for home security.

The integration of IoT technology further enhances the functionality of the system. With the help of NodeMCU and platforms like Blynk, users can monitor and control the system remotely through a mobile application. Real-time notifications and remote access provide greater convenience and flexibility, especially when users are away from home.

Overall, the Smart Home Security System is a practical, efficient, and scalable solution for modern homes. It reduces the need for manual monitoring, improves response time, and ensures better protection against intrusions. This project highlights the importance of sensor-based automation and IoT in developing smart and secure living environments, while also providing a foundation for future enhancements such as integration with cameras, alarms, and advanced wireless security systems.

10 Future Scope

The Smart Home Security System offers multiple opportunities for enhancement and can be further developed to make it more advanced, reliable, and intelligent:

- **Integration with AI and Machine Learning:** Future systems can use AI to differentiate between humans, animals, and objects, reducing false alarms and improving detection accuracy.
- **Camera Integration:** The system can be extended by adding surveillance cameras to capture images or videos when motion is detected, providing visual evidence of intrusions.
- **Mobile Notifications and Alerts:** Enhance the system to send instant push notifications, SMS, or email alerts to users when motion is detected, ensuring immediate awareness.
- **Biometric and Face Recognition:** Integrate face recognition or biometric systems to identify authorized users and detect unknown intruders more accurately.
- **Voice Assistant Integration:** Connect the system with voice assistants like Alexa or Google Assistant for hands-free control and monitoring.
- **Battery Backup and Power Optimization:** Add battery backup or low-power optimization techniques to ensure uninterrupted operation during power failures.
- **Multi-Sensor Integration:** Combine PIR sensors with additional sensors such as door sensors, gas sensors, or smoke detectors for a complete home security system.
- **Cloud Storage and Data Analytics:** Store event data in the cloud and analyze patterns to improve security strategies and detect unusual activities over time.

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